Block Two Synthesis. The Cholestatic Infant.

Picture a single week on a paediatric hepatology service. On Monday a breastfed girl of twenty-six days comes to clinic, gaining weight beautifully, with yellow eyes and stools her parents describe as yellow. On Tuesday a ten-day-old boy on formula is admitted to intensive care with a bloodstream infection caused by Escherichia coli, jaundice, and blood that will not clot. On Wednesday a three-week-old boy with a very small penis keeps dropping his blood sugar, and his liver tests look like a genetic disorder of bile transport. On Thursday a six-week-old girl with pale stools has a rapid genome result that comes back on the morning of her planned operation. And on Friday a boy born at twenty-nine weeks, now six weeks old after surgery for necrotising enterocolitis, has a rising conjugated bilirubin while still on intravenous nutrition.

All five infants share one sign: conjugated hyperbilirubinaemia, which is the laboratory word for cholestasis in the first months of life. That one sign has been the subject of the last thirteen lessons. This chapter brings the block together and asks one practical question. Which jaundiced infant cannot wait, and for what?

The main idea of the whole block can be said in one sentence. Neonatal cholestasis is a single sign shared by diseases that run on different clocks. Some clocks run in hours: bacterial sepsis, galactosaemia, and failure of the adrenal or pituitary glands can kill or injure the brain within days. One clock runs in weeks: biliary atresia, where every week of delay before surgery costs native liver survival. And some clocks run over months: nutrition, fat-soluble vitamins and growth, which fail slowly in every cholestatic infant whatever the cause. The work-up is therefore ordered by which clock runs fastest, not by which disease is commonest.

We will move through the material in five steps. First, we will lay out the mechanisms behind the sign. Second, we will compare the main documents, and look closely at where they disagree. Third, we will review the evidence that tells us what early diagnosis buys. Fourth, we will work through the five infants from the opening. Finally, we will collect twenty pearls, list what we still do not know, test ourselves, and decide what to do differently in clinic.

Part one. One sign, seven mechanisms.

It helps to think of every cause of neonatal cholestasis as one of seven mechanisms, because each mechanism has its own clue and its own first action.

The first mechanism is obstruction of the bile ducts. Biliary atresia, a progressive inflammatory destruction of the extrahepatic bile ducts, is the main example, and a choledochal cyst is the other. The clue is a pale stool, a high gamma-glutamyl transferase, called GGT, and a firm liver. The first action is a fasting ultrasound, followed by a liver biopsy or a cholangiogram, which is an X-ray picture of the bile ducts taken with contrast.

The second mechanism is inflammatory signalling. In sepsis and in urinary tract infection, bacterial endotoxin and inflammatory cytokines switch off the transporters that move bile acids and bilirubin out of the liver cell into the bile. The cholestasis is real, but it is transcriptional: it recovers when the infection is treated. Congenital cytomegalovirus, called CMV, syphilis, and herpes simplex virus belong here too. The important clue is that fever may be absent in a young infant. The first action is a urine culture, and in an infant with liver failure, acyclovir before the result of the herpes test.

The third mechanism is hormone deficiency. Hypopituitarism, hypothyroidism and isolated cortisol deficiency cause cholestasis, mainly because cortisol is needed for normal bile acid secretion. The clues are low blood sugar, a small penis in a boy, and midline defects of the face or brain. The first action is a cortisol and a free thyroxine level on the same day, and if both hormones are low, hydrocortisone must be given before thyroxine.

The fourth mechanism is a toxic metabolite. In galactosaemia, tyrosinaemia type one, citrin deficiency and, after weaning, hereditary fructose intolerance, a substance the infant cannot process injures the liver directly. The clue is a clotting disorder out of proportion to the jaundice, and in galactosaemia, reducing substances in the urine and infection with Escherichia coli. The first actions are to stop lactose on suspicion, to send the red cell enzyme test for galactosaemia before any transfusion, and to measure succinylacetone rather than tyrosine when tyrosinaemia is possible.

The fifth mechanism is a failure of bile transport or bile acid synthesis. Progressive familial intrahepatic cholestasis, called PFIC, Alagille syndrome, the bile acid synthesis defects and alpha-1 antitrypsin deficiency belong here. The clue in many is a low or normal GGT despite cholestasis. A low GGT with intense itch points to a transporter defect. A low GGT with low serum bile acids and no itch points to a synthesis defect. The first actions are to send urine bile acids by mass spectrometry before any ursodeoxycholic acid is started, and to send a genetic test early.

The sixth mechanism is infiltration or alloimmune injury. Haemophagocytic lymphohistiocytosis, called HLH, Niemann-Pick disease type C, Wolman disease, which is the infant form of lysosomal acid lipase deficiency, and gestational alloimmune liver disease, called GALD, belong here. The clues are a large spleen, a very high ferritin, and liver failure that begins at birth. The first actions are to apply the HLH diagnostic criteria, to look for iron outside the liver on a lip biopsy or MRI in suspected GALD, and in GALD to give intravenous immunoglobulin with exchange transfusion.

The seventh mechanism is iatrogenic and preterm cholestasis, mainly parenteral nutrition associated cholestasis, often combined with sepsis, necrotising enterocolitis and bowel surgery. The clue is a clear dose-response with days of intravenous nutrition. The first action is to advance enteral feeds and change the lipid emulsion, while still excluding atresia.

Part two. The documents, and where they part.

Five documents anchor this block. The first is the 2017 joint guideline of the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, called NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, called ESPGHAN, written by Fawaz and colleagues. It uses the GRADE method and gives ten numbered recommendations, most supported by low-quality evidence. Its line to remember is that a direct or conjugated bilirubin above one milligram per decilitre is abnormal whatever the total bilirubin, and that infants should be tested at two weeks of age, or at three weeks if breastfed and otherwise well.

The second is the 2022 clinical practice guideline of the American Academy of Pediatrics, called the AAP, on hyperbilirubinaemia in infants born at thirty-five weeks or later, written by Kemper and colleagues. It is mainly about unconjugated jaundice and phototherapy, but it contains a key instruction about cholestasis: a breastfed infant who is still jaundiced at three to four weeks, or a formula-fed infant who is jaundiced at two weeks, should have total and direct bilirubin measured.

The third is the 2025 AAP clinical report for primary care on identifying biliary atresia, written by Harpavat and colleagues. Its central message is that any newborn direct or conjugated bilirubin above the laboratory reference range is abnormal, even if only slightly, and that infants with atresia should be identified between two and four weeks of life, so that surgery can happen before about thirty days.

The fourth is not a guideline but a set of programmes: the stool colour cards used nationally in Taiwan since 2004, in Japan since 1994 and nationally since 2012, and in the Canadian province of British Columbia since 2014. Their lesson is that the same card can be highly or poorly sensitive depending on the programme around it.

The fifth is the British guideline from the National Institute for Health and Care Excellence, called NICE, on neonatal jaundice, which defines prolonged jaundice as more than fourteen days at term and more than twenty-one days in preterm infants, and treats a conjugated bilirubin above twenty-five micromoles per litre as significant.

Now to the disagreements. On the definition of an abnormal value, the documents differ. The joint 2017 guideline uses one milligram per decilitre and has dropped the old rule that the direct fraction must exceed twenty percent of the total. The 2025 AAP report goes further for the newborn period and calls any value above the laboratory range abnormal. NICE uses twenty-five micromoles per litre, which is slightly higher, and an Italian society guideline from 2022 keeps the twenty percent rule.

On the timing for formula-fed infants, the documents are aligned: test a formula-fed infant who is jaundiced at two weeks.

On the timing for breastfed infants, the difference is material. The joint guideline says three weeks if the infant is well, the AAP 2022 guideline says three to four weeks, and NICE says fourteen days for every infant whatever the feeding. This matters more than it seems. If the target is surgery before thirty days, an infant whose first fractionated bilirubin is drawn at four weeks has already used up almost the whole window. In that infant, the early signal has to come from somewhere else: from the stool, and from any bilirubin measured in the first days of life.

On the stool, the documents are aligned in spirit. A pale stool at any age is a reason to test that day. The AAP asks primary care to ask about stool colour at every visit between two and four weeks, and Taiwan and Japan put a printed card in every child health booklet.

On universal screening, they differ. The AAP report asks that a direct bilirubin be added to every bilirubin draw in the newborn period. The United Kingdom National Screening Committee does not recommend universal screening, and the joint 2017 guideline does not address it.

On the target age for the Kasai operation, the difference is again material. North America now names a number, before thirty days. European documents ask for speed without a number, and British national data show a median age close to fifty days, which has been judged acceptable there.

On the radionuclide scan of bile excretion, called HIDA, the documents differ. The joint guideline says excretion of tracer into the bowel helps to exclude atresia, while the Italian guideline no longer includes the scan in the standard work-up. And for preterm infants there is a gap: the joint guideline has no separate timing, the AAP guideline excludes infants born before thirty-five weeks, and only NICE gives a preterm threshold of twenty-one days.

Part three. What early diagnosis buys.

Why does the timing argument matter so much? Because the age at Kasai portoenterostomy predicts whether the child keeps their own liver. In a Canadian national cohort of three hundred and forty-nine infants, survival with the native liver at four years was forty-nine percent when the Kasai was done by thirty days of age, thirty-six percent between thirty-one and ninety days, and twenty-three percent after ninety days. In the French national cohort, native liver survival fell step by step with every fifteen-day band of age at surgery.

Can screening move the date? The strongest evidence comes from Texas, where Harpavat and colleagues measured direct or conjugated bilirubin twice in more than one hundred and twenty-four thousand newborns: once before sixty hours of life on the routine sample, and again by two weeks if the first value was above the hospital's reference range. The programme found all seven infants with atresia, with a specificity of ninety-nine point nine percent, although the positive predictive value was only about six percent. The centre then compared its atresia patients before and after screening. Before screening, twelve and a half percent had

their Kasai by thirty days; after screening, fifty-eight percent did. Normal bilirubin by ninety days after surgery rose from forty-two to seventy-nine percent. Survival without transplant at one year rose from seventy-one to ninety-five percent, but this last difference did not quite reach statistical significance, with a p value of point zero six. The mean age at Kasai fell from fifty-six to thirty-six days. Seven cases make a small study, and the lower bound of the confidence interval for sensitivity is about fifty-six percent, but the direction is clear.

The stool card tells a different story. Its sensitivity depends on the programme. In the second year of Taiwan's national programme it reached ninety-seven percent, and in the regional pilot before that, about ninety percent. A meta-analysis of seven studies and almost one million infants found a pooled sensitivity of about eighty percent. In Tochigi in Japan, where the card was checked once at the one-month visit, sensitivity was about seventy-seven percent. In British Columbia, where the card was handed out passively, it found three of six infants with atresia, a sensitivity of fifty percent. And in a Sapporo cohort, where the card was checked once at one month, both infants with atresia still had normal-looking stools at that visit. Why such variation? Taiwan has about three times the prevalence of atresia, the card is read repeatedly rather than once, and an abnormal colour had to be reported within twenty-four hours. In Taiwan, five-year jaundice-free survival with the native liver rose from twenty-seven to sixty-four percent after the card was introduced, but that was a before-and-after comparison over a decade in which surgery and transplant care also improved. The practical rule is simple: a normal card is not reassurance in a jaundiced infant, and a pale stool is a same-day referral.

What do jaundiced infants turn out to have? A systematic review of one thousand six hundred and ninety-two infants with conjugated hyperbilirubinaemia, published by Gottesman and colleagues in 2015, found idiopathic neonatal hepatitis in twenty-six percent and biliary atresia in almost exactly the same proportion. Infection explained eleven and a half percent, parenteral nutrition six percent, metabolic disease about four percent, alpha-1 antitrypsin deficiency about four percent, and hypopituitarism or hypothyroidism two percent. So atresia is one in four, and the causes that kill or maim within days if missed, namely infection, metabolic and endocrine disease, add up to about one in six. The idiopathic group is shrinking as genetic testing spreads; modern algorithms leave perhaps fourteen percent or fewer without a name, and gene panels find a cause in up to seventy percent of infants whose features look genetic.

Six further numbers from the block deserve to be carried into clinic. First, seventy-nine percent of infants with atresia had a direct-to-total ratio of twenty percent or less on the second day of life, so the old ratio rule would have called them normal. Second, in the 1977 Massachusetts cohort of galactosaemia, four of ten infants had Escherichia coli infection and three died. Third, six months after a Kasai, none of twenty-three infants whose bilirubin was still two milligrams per decilitre or more was sufficient in all four fat-soluble vitamins on a combined multivitamin product. Fourth, among infants with Alagille syndrome who had a Kasai, forty-seven percent went on to transplant, compared with fourteen percent of infants with atresia, and thirty-two percent died, compared with three percent. Fifth, in the syndrome of

arthrogryposis, renal dysfunction and cholestasis, called ARC, seven of sixteen children who had a liver biopsy bled dangerously, mostly with normal platelet counts. Sixth, in gestational alloimmune liver disease, weekly intravenous immunoglobulin during the next pregnancy reduced the recurrence of an affected infant from about forty-five percent to about five percent.

Part four. Five difficult infants.

Return now to Monday's infant: a breastfed girl of twenty-six days, thriving, jaundiced since day four, with stools her parents call yellow and that match colour four on the card. Her day-two bilirubin was nine point eight milligrams per decilitre in total with a direct fraction of point six, when the laboratory's upper limit was point three, and nobody repeated it. Today her total bilirubin is seven point nine and her direct bilirubin is two point one. How should we read this? A direct bilirubin of two point one is cholestasis whatever the total, and thriving and breast milk explain nothing. The day-two value above the laboratory range was the first signal, because in atresia the conjugated bilirubin is abnormal from birth. Stools can remain pigmented early in atresia, so a normal card does not exclude it. On the first day she needs a urine culture and urine reducing substances, a glucose, an international normalised ratio, called INR, a thyroid stimulating hormone with free thyroxine, and parenteral vitamin K. The decision is a same-week hepatology review with a fasting ultrasound and then biopsy or cholangiogram, aiming for a Kasai, if it is needed, well before the end of the sixth week.

Tuesday's infant is a ten-day-old boy on standard formula who has fed poorly and vomited since day five. His blood culture grows Escherichia coli. His direct bilirubin is four point one, his alanine aminotransferase is one hundred and forty, his INR is three point two and his glucose is low. Cataracts are not yet visible and the newborn screen is pending. Gram-negative sepsis together with cholestasis and a clotting disorder means galactosaemia until proven otherwise. Lactose should be stopped today, before any result, because the diet is both diagnostic and therapeutic. The red cell enzyme assay, galactose-1-phosphate uridyltransferase, must be sent before any transfusion, otherwise donor red cells make the result falsely normal for months. An INR that is out of proportion to the bilirubin should also raise tyrosinaemia, and the right test is succinylacetone, not tyrosine. So he receives antibiotics, a lactose-free formula, parenteral vitamin K and glucose immediately, and the diagnostic samples are drawn before any blood products.

Wednesday's infant is a three-week-old term boy with recurrent blood glucose between two point one and two point six millimoles per litre and a micropenis. His direct bilirubin is three point six, his GGT is only sixty, and his thyroid stimulating hormone is normal, as was his newborn screen. A genetic panel for low-GGT cholestasis has already been sent. In published series, low blood sugar was present in fourteen of sixteen cholestatic infants with hypopituitarism; it is the single best bedside clue. GGT is normal in about half of these infants, so endocrine cholestasis can mimic PFIC. A normal thyroid stimulating hormone does not exclude central hypothyroidism, because the problem is in the pituitary, so free thyroxine must

be measured. A random cortisol below about five point six micrograms per decilitre is specific, but a normal value excludes nothing. The decision is to measure cortisol and free thyroxine today, to give hydrocortisone before levothyroxine so as not to provoke an adrenal crisis, and to image the pituitary. The jaundice should clear over two to six months once hormones are replaced; the danger to life is adrenal crisis, not the liver.

Thursday's infant is a six-week-old girl with acholic stools, a direct bilirubin of five point eight and a GGT of eight hundred and twenty. Her ultrasound shows a small gallbladder without the triangular cord sign. Her HIDA scan shows no excretion at twenty-four hours. Her biopsy shows ductular reaction and bile plugs, with only four portal tracts. An echocardiogram shows peripheral pulmonary stenosis. On the morning of the planned cholangiogram, the rapid genome returns a pathogenic frameshift variant in the gene JAG1. Infants with Alagille syndrome can have pale stools and a non-excreting HIDA scan, and ductular reaction on biopsy can mimic obstruction. This matters, because when a Kasai is done in Alagille syndrome the outcome is worse than in atresia, with transplant in almost half and death in about a third. The order of work is to look for the features of Alagille syndrome first, with an echocardiogram, a spine film and an eye examination, and to perform a cholangiogram only if doubt about the duct remains. If the duct is patent, there should be no portoenterostomy. Supportive care starts at once, and itch can later be treated with an ileal bile acid transporter inhibitor. The general rule is worth memorising: a genetic result may cancel a Kasai, but a pending genetic result must never delay a cholangiogram.

Friday's infant was born at twenty-nine weeks and is now thirty-five weeks corrected, after six weeks of parenteral nutrition following a bowel resection for necrotising enterocolitis. His direct bilirubin is four point four and rising, his GGT is three hundred and ten, enteral feeds are only thirty millilitres per kilogram per day, and the nurses have seen pale stools on two days this week. Parenteral nutrition associated cholestasis is common: about forty-three percent after four to eight weeks of intravenous nutrition, and seventy-two to eighty-five percent beyond eight weeks. But common is not exclusive. Preterm infants with atresia had their Kasai at a mean of seventy-two days compared with fifty-three days at term, largely because cholestasis was blamed on intravenous nutrition. In this setting HIDA scans and biopsies are confounded, so stool colour and the trajectory of the bilirubin decide. Fish oil lipid at one gram per kilogram per day rescues established parenteral nutrition associated cholestasis, and enteral feeding is the real cure. The plan is to change the lipid, advance feeds, and check the stool card every day. If pale stools persist, or the bilirubin keeps rising despite feeds, he needs an ultrasound and a cholangiogram.

Part five. Twenty pearls, the gaps, and a self-test.

Here are twenty pearls from the block. Neonatal cholestasis occurs in about one in two thousand five hundred term births, and atresia and idiopathic hepatitis each make up about a quarter. A direct bilirubin above one milligram per decilitre is abnormal whatever the total, and

the old twenty percent rule missed seventy-nine percent of atresia on day two. Fractionate the bilirubin at two weeks in a formula-fed infant, at three weeks in a breastfed and well infant, and on the same day at any age when the stool is pale. A Kasai by thirty days gives about a forty-nine percent chance of keeping the native liver at four years, and after ninety days about twenty-three percent. Two-stage blood screening in Texas found seven of seven cases and moved the mean age at Kasai from fifty-six to thirty-six days. Stool card sensitivity runs from fifty percent in British Columbia to ninety-seven percent in Taiwan, so the programme decides. Send a urine culture on the first day, and treat Escherichia coli infection in a cholestatic infant as galactosaemia until excluded. Low blood sugar plus cholestasis means cortisol and free thyroxine today, with hydrocortisone first. Congenital CMV of moderate to severe degree is treated with six months of valganciclovir, and it is never a reason to delay a cholangiogram. In neonatal liver failure, start acyclovir before the herpes test returns, and an INR out of proportion to the jaundice suggests tyrosinaemia or GALD.

The second ten continue the story. A low GGT and low serum bile acids without itch suggest a bile acid synthesis defect, so send urine bile acids before any ursodeoxycholic acid. Citrin deficiency in the newborn shows a high GGT, a very high alpha-fetoprotein and a high citrulline, and almost all infants recover by twelve months on a lactose-free formula rich in medium-chain triglycerides, called MCT. Parenteral nutrition cholestasis is dose dependent, rising from about fourteen percent to about eighty-five percent with longer exposure; enteral feeds cure it and fish oil rescues it. Sporadic idiopathic hepatitis with pigmented stools usually recovers, while familial cases are genetic until proven otherwise. A large spleen in a cholestatic infant suggests storage disease or HLH, and a ferritin above ten thousand is about ninety percent sensitive for HLH. Never biopsy an infant in whom ARC syndrome is suspected. Genetic yield runs from about four percent in parenteral nutrition cholestasis to seventy-one percent when the features look genetic, and in one trial genome sequencing diagnosed forty-nine percent of infants compared with twenty-seven percent by targeted panel. In Alagille syndrome a Kasai leads to transplant in forty-seven percent compared with fourteen percent in atresia, so the echocardiogram comes first and the cholangiogram second. After a Kasai, if the bilirubin stays at two or above, dose each fat-soluble vitamin separately and measure each. And at six months after Kasai, a weight-for-age z-score of minus two point one compared with minus one point two separated children with a poor outcome from those with a good one; measure the mid-upper arm, not only the weight, because liver, spleen and fluid add weight that is not nutrition.

What could the block not answer? There is no single agreed timing for fractionating bilirubin in a breastfed infant; we have three, two, and three to four weeks in three documents, and no cohort has linked the age at first direct bilirubin to the age at Kasai. Universal direct bilirubin screening rests on only seven cases in the largest American study, and needs a multistate programme with audited recall. There are no validated direct bilirubin cut-offs or timings below thirty-five weeks. Ursodeoxycholic acid after a Kasai is given almost universally but has never been tested against placebo; a randomised withdrawal trial would answer it. Giant cell

hepatitis with autoimmune haemolytic anaemia has no guideline, and mortality has ranged from about ten to forty-five percent across eras. The neonatal guideline does not tell us to arrange antenatal immunoglobulin for the next pregnancy after GALD, even though it works. And no document says explicitly that a variant of uncertain significance, or a genetic test still pending, must never delay the cholangiogram.

Test yourself with ten questions. A newborn on day two has a direct bilirubin of point five with a laboratory limit of point three: is that abnormal? Yes, according to the 2025 AAP report; repeat it by two weeks and ask about stools. A breastfed, thriving, jaundiced infant at three weeks: which test? Total and direct bilirubin, whatever the stool colour. A cholestatic infant with Escherichia coli sepsis: what diet change? Stop lactose now, and send the enzyme assay before any transfusion. Low GGT, low sugar and a normal thyroid stimulating hormone: which tests? Cortisol and free thyroxine, and hydrocortisone before thyroxine. A tyrosinaemia screen that measured tyrosine was normal: is tyrosinaemia excluded? No; send succinylacetone. Low GGT, low serum bile acids and no itch: what before ursodeoxycholic acid? Urine bile acids by mass spectrometry. A preterm infant on intravenous nutrition has pale stools: can we blame the nutrition? No; ultrasound, a daily stool card, and a cholangiogram if the pallor persists. A pathogenic JAG1 variant with a non-excreting HIDA scan: should we do a Kasai? Not if the duct is patent; echocardiogram first. Suspected ARC syndrome: biopsy for diagnosis? No; genetics, because seven of sixteen biopsied children bled. And the mother of an infant who died of GALD is pregnant again: what now? Weekly antenatal immunoglobulin from mid-pregnancy, which reduces recurrence from about forty-five to about five percent.

What to do differently in clinic.

So what changes tomorrow? Look up the newborn direct bilirubin of every jaundiced infant, and treat a value above the laboratory range as a signal rather than a footnote. Fractionate the bilirubin of every jaundiced breastfed infant by three weeks, and of any infant with a pale stool on the day the stool is seen. Before starting the atresia work-up, run the day-one list: urine culture, urine reducing substances, glucose, INR, free thyroxine, cortisol, and parenteral vitamin K. Stop lactose on suspicion of galactosaemia, send the enzyme assay before blood, and measure succinylacetone rather than tyrosine. Look at the stool yourself at every visit, and teach the parents to use the card. Ask for a rapid genome in parallel with the atresia work-up, and never let it delay the cholangiogram. Send urine bile acids before the first dose of ursodeoxycholic acid. Dose each fat-soluble vitamin separately, measure each, and follow the arm circumference rather than the weight alone. And after GALD, or after familial idiopathic hepatitis, refer the family to genetics and obstetrics, because the next pregnancy is also our patient.